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A Novel Laser-Based Method for Stimulating Wellbore Production

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Abstract

Lack of permeability can hinder the flow of hydrocarbons from the formation to a wellbore. This method utilizes a high-power laser technology combined with enablers to effectively stimulate the wellbore for production. It consists of a conduit, a drill head, and an optical fiber that delivers the laser beam. The activated agent then increases the energy absorbed by the formation, which makes the laser stimulation more effective.

By combining a powerful laser with an activated agent, the new method can stimulate the flow of hydrocarbons and fracture the formation. The first step involves placing the drill head into the wellbore, which is equipped with an activated agent and optical fiber for delivery of the laser. The activated agent is then discharged onto the formation using a nozzle. The laser beam then passes through the optical fiber and bathes the agent-coated region. The ability to maneuver the drill head allows the laser and the agent to precisely target certain areas of the formation.

The combination of the laser and an activated agent provides a unique opportunity to enhance the well's production and reduce the environmental impact. This method offers numerous advantages over traditional techniques, such as hydraulic fracturing. The ability to combine the two components marks a significant advance in the field of well stimulation, and it paves the way toward a future where this technology could become a game-changer. Its potential to increase production, lower environmental impact, and offer more control over the process makes it an appealing technology. The development of the laser technology has led to revolutionary approach in the field of well stimulation. It could help unlock new reserves and significantly reduce the environmental impact of the process.

This method allows engineers to achieve a precise and controlled fracturing while reducing the risk of encountering undesirable formations. The laser-based stimulation can result in increased production and efficiency. It also offers theoretical advantages, such as lower resource consumption and better flowrates. This opens up a wealth of possibilities well stimulation. Further studies and testing are needed to confirm its full potential.

Introduction

Well stimulation is a crucial step to sustain well production, as it enhances communication between the wellbore and the formation, allowing hydrocarbons to flow and produce. Such flow can be challenging, especially in unconventional reservoirs, including tight formations, where permeability is low. Various technologies have been employed to stimulate the wellbore, but they come with their own set of challenges. For instance, hydraulic fracturing, a technology used to create channels in the formation by applying high-pressure fluid to crack the rock, poses several challenges, including the use of fresh water, chemicals, large surface footprints, and the requirement of multiple high-pressure pumps. Another significant challenge associated with hydraulic fracturing is the breaking pressure of the rocks, which can be extremely high, necessitating the need for pumps at the surface to pump fluid at very high rates.

This paper introduces a field-proven technology that has already been successfully deployed in the field, utilizing high-power laser technology. The advantage of this technology lies in its non-damaging, waterless, and flow-enhancing capabilities. Field deployments have demonstrated that this technology is not only effective but also environmentally friendly, safe, and clean, as the power source is mounted on the surface and the power is delivered via a fiber-optic cable to the wellbore.

Laser Technology

Laser technology possesses unique properties that make it a candidate for downhole applications. Specifically, laser is stress and formation independent, it is proven that the beam has the ability to penetrate all rock types regardless of their type and compressive strength (Batarseh et al. 2017). For example, laser has successfully penetrated extremely strong rocks, such as basalt, with compressive strengths of up to 42,000 psi. The holes created by laser are clean and symmetrical, without any deviation. Additionally, the high-power laser (HPL) beam is heterogeneity independent, as demonstrated through several lab experiments that have shown the beam to penetrate highly laminated unconventional rock samples with high precision (Batarseh et al 2019a). Fig. 1 shows images of a CT scanned sample of highly laminated shale before and after laser perforation. The HPL laser beam created a straight cut through the different shale layers, independent of heterogeneity. Moreover, the presence of a geologic intrusion (e.g. fault) does not affect the penetration of the HPL beam. Fig. 2 illustrates a photo of a HPL perforated sample. The perforation continued to penetrate deep through the sample despite the presence of a calcite intrusion.

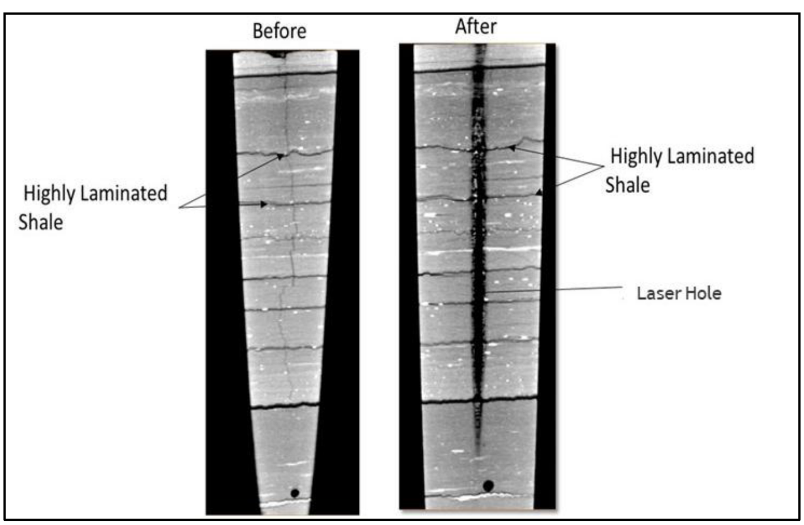

Figure 1—CT images of a HPL perforation through a highly laminated shale formation (Batarseh et al. 2019a). The penetration shows how the process is unaffected by heterogeneity, as the HPL creates a straight hole while the bulk of the rock remains intact.

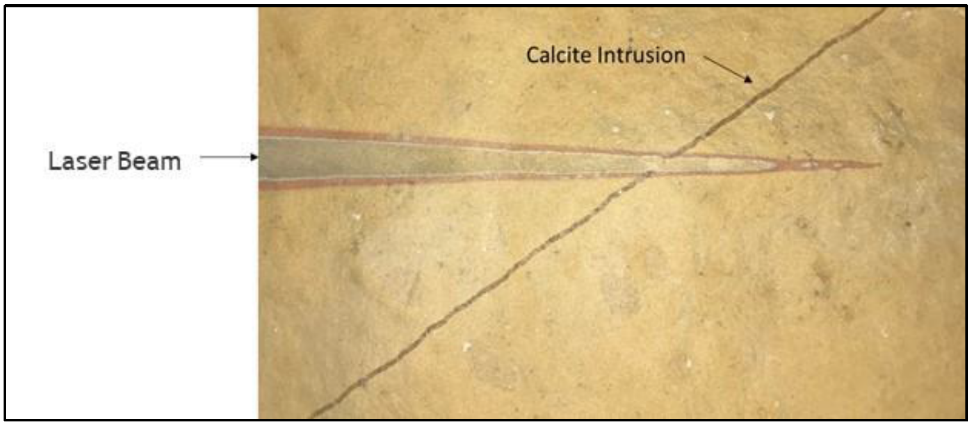

Figure 2—High-power laser perforation of a rock with an intrusion (Batarseh et al. 2019a). The photo shows how HPL can be controlled and directed to create a straight hole in the formation regardless of heterogeneity.

The phase of the rock can also be controlled by manipulating the laser beam. The heat created by the HPL beam can provide the energy to spall, melt, or sublime the rock, depending on the thermal and electromagnetic properties of the rock (Batarseh et al. 2019b). In addition, the high-power laser technology has demonstrated in the lab the capability to create fractures in the rock. Fig. 3 depicts laser perforations through different rock types ranging from Berea sandstone to granite. This indicates the versatility and effectiveness of the laser technology in various downhole applications.

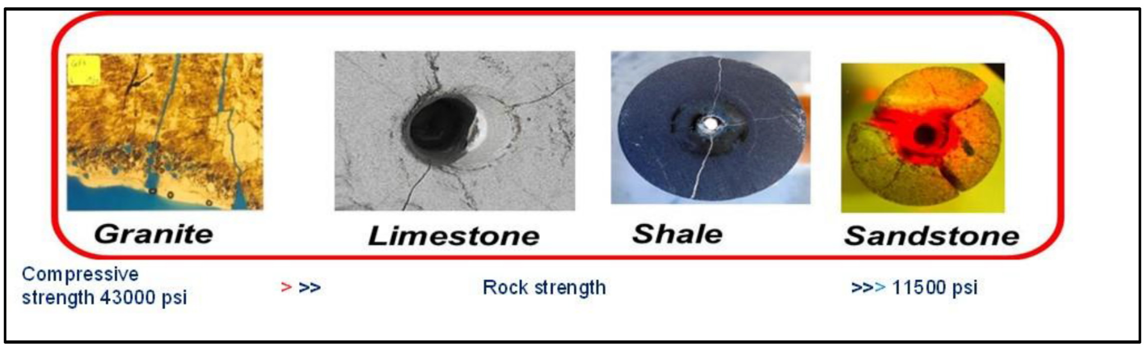

Figure 3—High-power laser fracturing initiation in granite, limestone, shale, and sandstone samples despite the great variance of compressive strengths among the tested samples (Batarseh 2001).

Various experimental tests reported in the literature showed improvements in porosity and permeability measurements as a result of laser perforation. Batarseh et al. (2017) evaluated the effect of high-power laser on porosity and permeability by testing different rock types. Table 1 presents the results of their experiments, indicating the percentage increase in porosity and permeability measured values across 6 rock samples.

Table 1—Percentage Increase in Permeability and Porosity Measurements Post Laser Perforation (Batarseh et al. 2017)

Sample	Permeability increase %	Porosity increase %
Berea Yellow	2	57
Berea Gray	22	50
Reservoir Sandstone	171	150
Limestone	33	15
Shale 1	28	700
Shale 2	11	250

Laser Field Applications

The unique features and advantages of lasers, discussed in the previous section, can be harnessed to solve multiple challenges in the oil and gas industry. High-power lasers have been deployed in oil and gas fields mainly for two applications: flowline descaling, and well perforation. Batarseh et al. (2017) presented the structure of a high-power laser field system. The system consists mainly of three key elements: the surface system, which handles laser generation and includes various ancillary equipment; the fiber optic cable, which can be injected in a coiled tubing and is responsible for transmitting the laser beam from surface to target; and the bottomhole assembly, which mainly comprises an optical system (mirrors, lenses, etc.), a rotational assembly (allowing the rotation of the laser head), and a purging system to ensure efficient beam delivery with minimal absorption from the downhole fluids. Fig. 4 illustrates the schematic of the high-power laser bottomhole assembly.

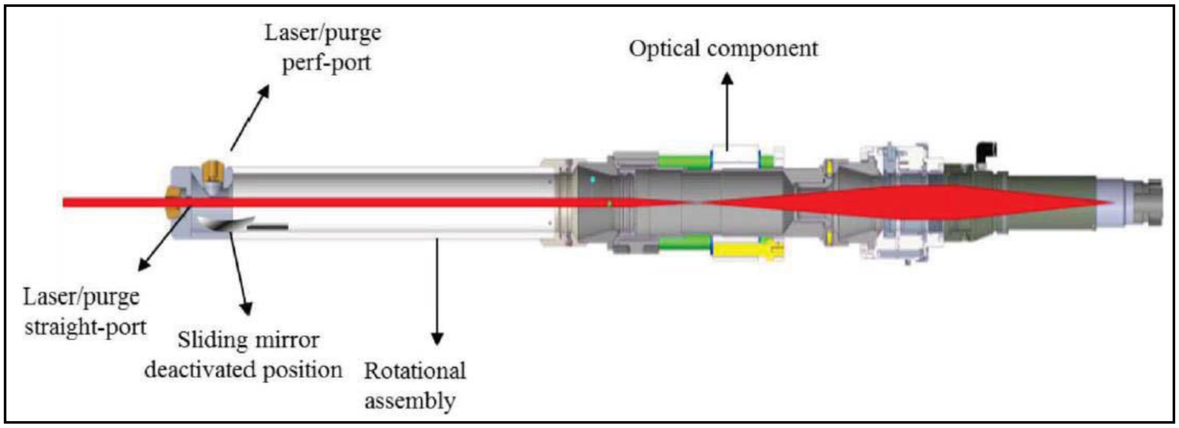

Figure 4—Schematic of the bottomhole assembly of a high-power laser perforation field system (Batarseh et al. 2017).

Novel Method for Improving the Efficiency of Laser Stimulation

High power laser has proven to be effective in downhole application, it is a controlled heat source that can be focused precisely on the target. The beam shape and size can be controlled precisely by manipulating the optics, beam can be straight with fixed diameter (collimated beam) or focused creating conical shape. The mechanism of the laser interaction with the formation is by creating heat. This work introduces a novel approach to improve the efficiency of the operation using the same laser source, this can be done by utilizing a compact effective tool that is used for heating the wellbore combined with enabler materials (Alshunaifi et al. 2022). The uniqueness of this technology is that when the enablers are exposed to the laser energy, they heat up instantly reaching high temperature in the wellbore (or surface) in seconds, this causes the temperature in the rock to elevate reaching up to 1795 °C. This high energy can be utilized in a tool that heats up for several applications such as reducing fracturing break pressure, heating, and to generate in-situ steam.

When exposing rock/materials to the laser beam, the energy from the beam to the rocks/materials will be absorbed, the amount of energy absorbed by the rock sample from the laser depends on many factors. The main factor is the color of the sample, the brighter the color, more reflection will occur, causing a loss of energy. To overcome this challenge, improve the efficiency of the laser-rock energy transfer, and ensure consistency of the energy in the rocks, a proof of concept experimental work has been conducted on activated carbon (AC). The lab tests found that the laser interaction with the activated carbon increases the temperature of the sample from 880 to 1790 °C. This sudden increase in temperature results in thermal shocking in the formation and rock creating micro fractures, as shown in Fig. 5. This is one of the applications that demonstrate the potential of high-power laser on rocks.

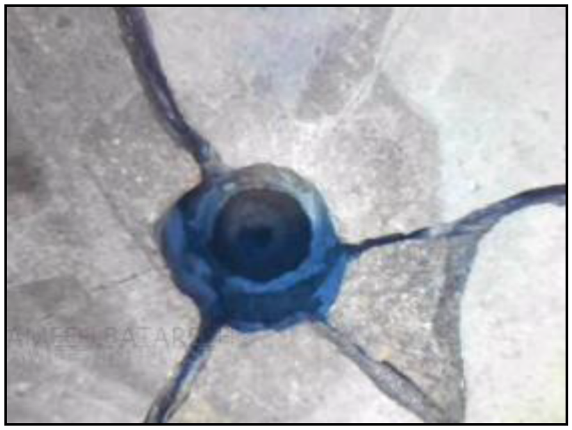

Figure 5—Thermally fractured rock by high-power laser.

A series of laboratory-scale tests has been performed to evaluate different scenarios. A laser beam of 1 kW was applied to three sample types, as per the following scenarios:

- **Scenario 1:** Laser stimulation of dry limestone rock sample (without activated carbon)
- **Scenario 2:** Laser stimulation of dry limestone rock sample (with activated carbon)
- **Scenario 3:** Laser stimulation of water-wet limestone rock sample (with activated carbon)

Fig. 6 shows the experimental setup for Scenario 1 and Scenario 2, where one area of the dry sample is covered with activated carbon (AC), and another area without. An infrared (IR) camera captured the temperature of the heated spot while heating lasted for 30 seconds for the three scenarios. Fig. 7 shows an IR camera image of the laser heated limestone sample without activated carbon. The maximum temperature recorded by the IR camera was 888 °C, as indicated in the temperature plot in Fig. 8.

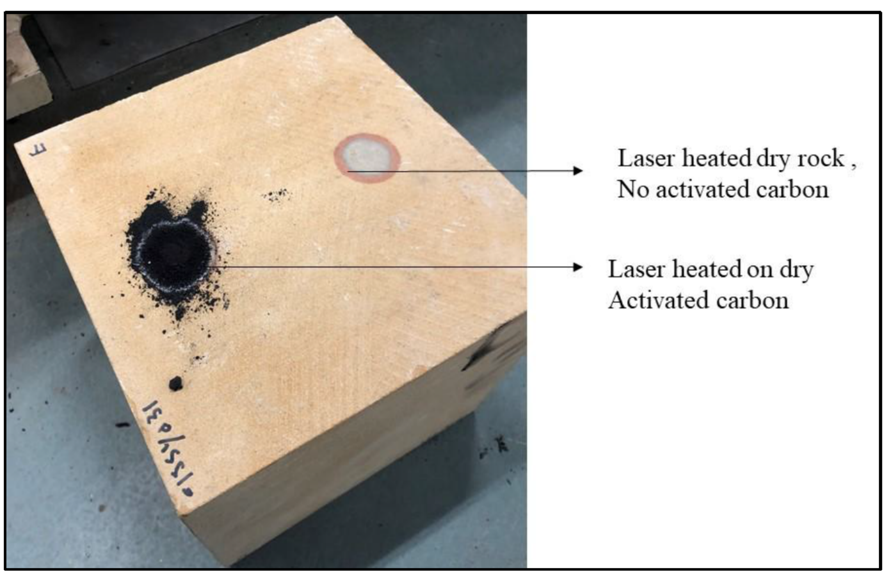

Figure 6—Experimental setup showing the limestone rock sample with one area covered with AC and another without AC.

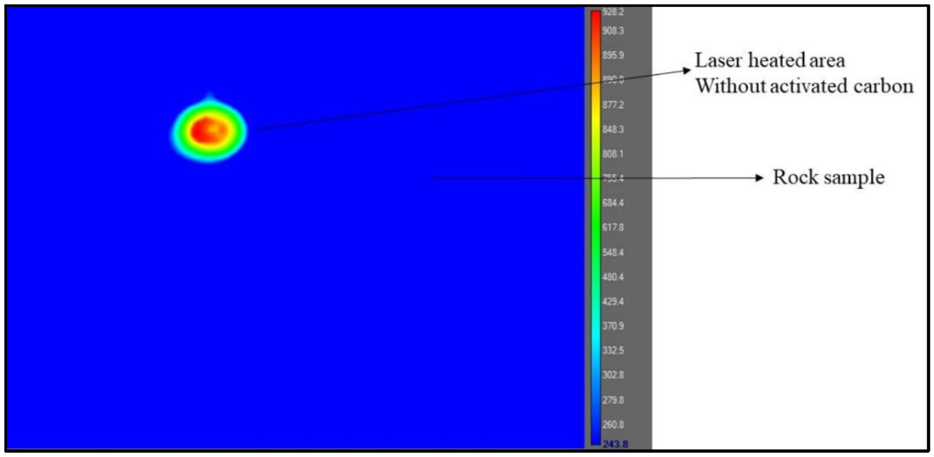

Figure 7—IR image of laser heated dry rock sample without activated carbon (Scenario 1).

Figure 8—Temperature plot from the IR camera of the laser heated dry rock sample without activated carbon (Scenario 1).

The laser head was then moved to the area that is covered with activated carbon for Scenario 2 test. The area was heated for the same amount of time, 30 seconds. Fig. 9 shows an IR image of the laser heated rock sample while the laser is focused on the activated carbon. The maximum temperature value recorded was 1795 °C, as illustrated in Fig. 10.

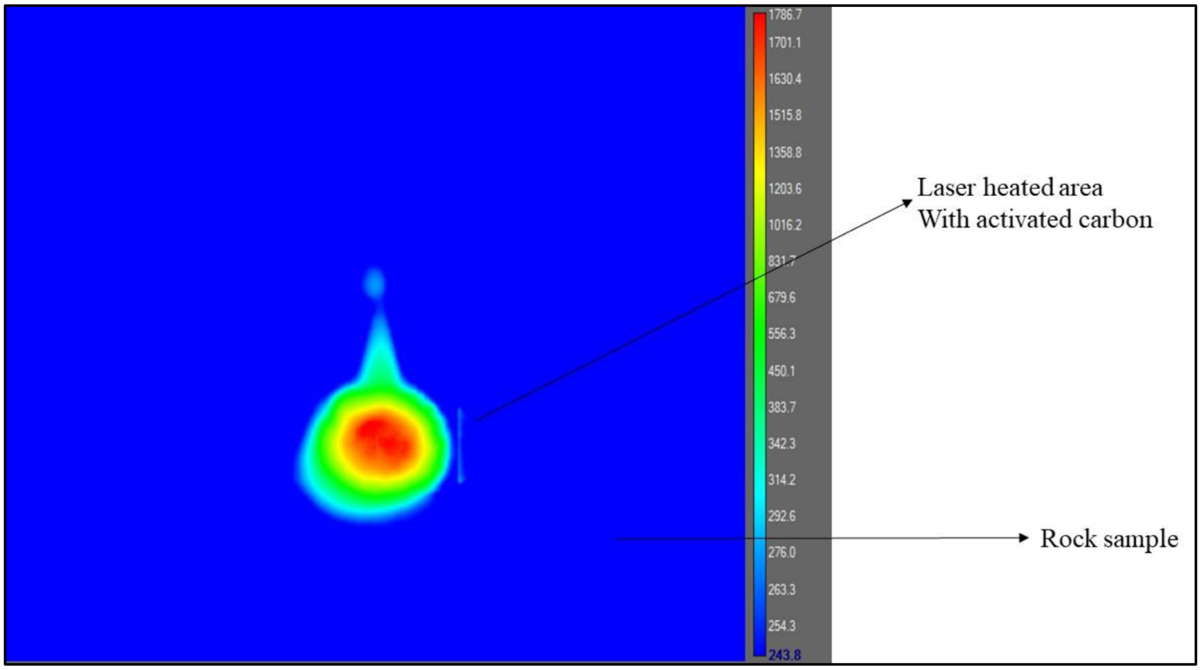

Figure 9—IR image of laser heated rock sample with laser focused on the activated carbon (Scenario 2).

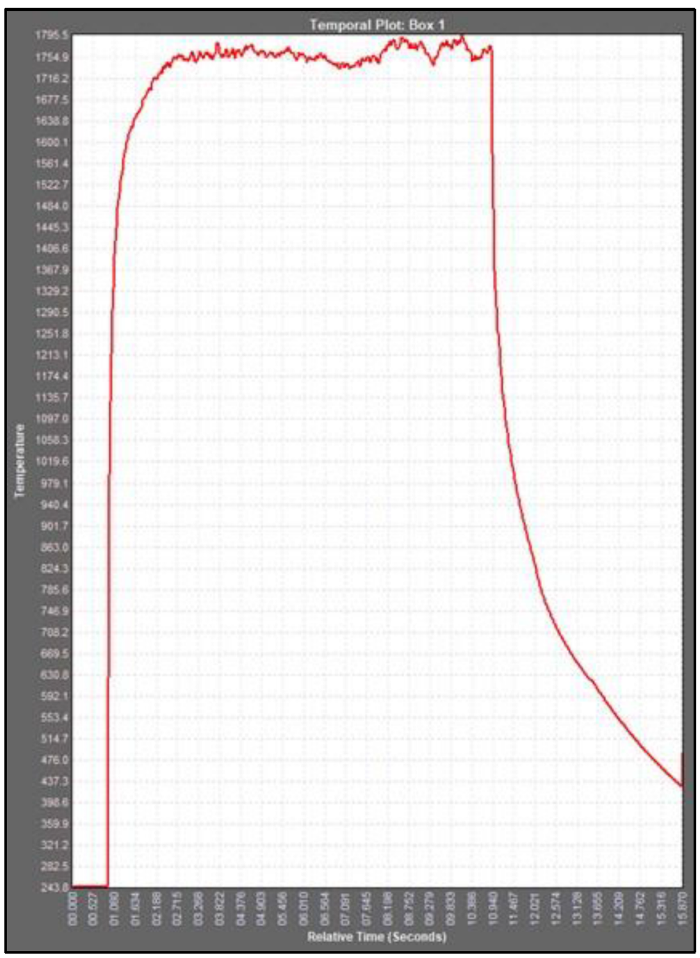

Figure 10—Temperature plot from the IR camera of the laser heated rock sample with laser focused one the activated carbon (Scenario 2).

For Scenario 3, the sample was then saturated with water with the activated carbon being water wet to simulate downhole conditions. Fig. 11 demonstrates the experimental setup after saturating the sample with water. The experiment was repeated and an IR image was captured, as shown in Fig. 12. The maximum temperature recorded when the laser was focused on the wet activated carbon was 1564 °C, as indicated in Fig. 13. The results confirm that in both cases, wet (Scenario 3) or dry rock sample (Scenario 2), the presence of activated carbon increases the temperature of the laser heated sample by at least 2 times. This method improves heat transfer to the rock even if the laser power is low, as this example uses 1 kW only.

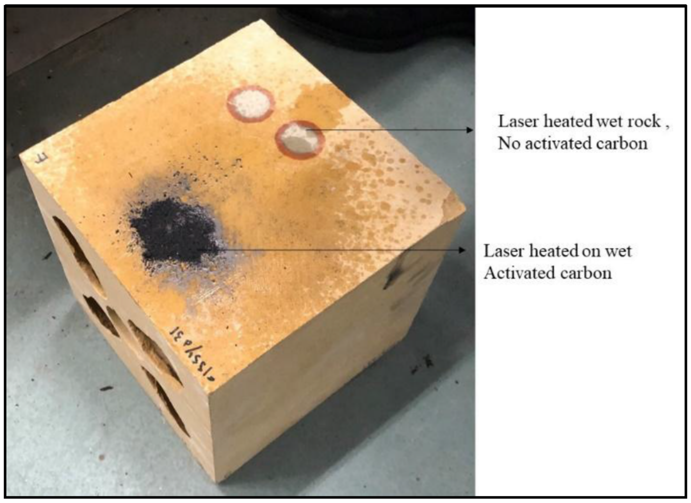

Figure 11—Experimental setup showing the limestone rock sample with one area covered with water-saturated AC and another without AC.

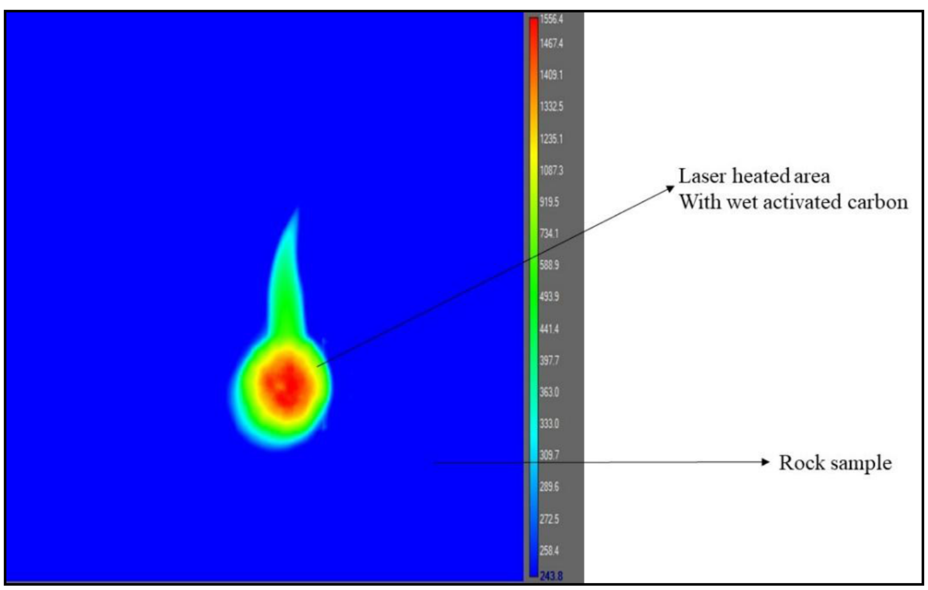

Figure 12—IR image of laser heated rock sample with laser focused on the water-wet activated carbon (Scenario 3).

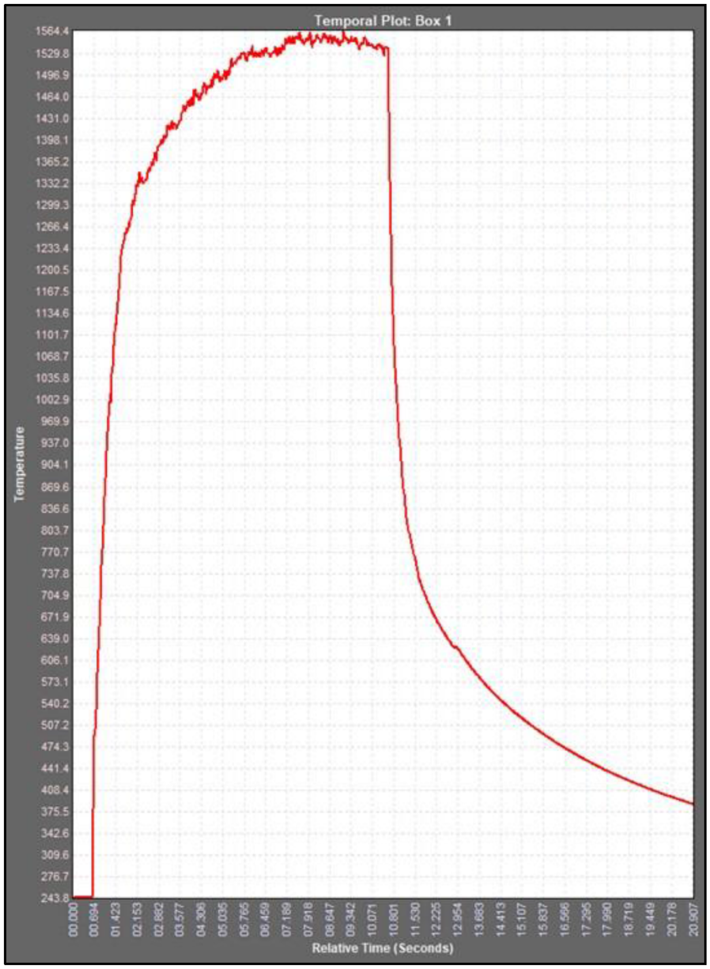

Figure 13—Temperature plot from the IR camera of the laser heated rock sample with laser focused one the water-wet activated carbon (Scenario 3).

The results of the water-saturated sample (Scenario 3) indicate that water absorbs the laser energy and reduces the heat transfer to the formation. However, the temperature was still very high at 1564 °C. This proves that when the activated carbon is in contact with the wet surface of the rock, the generated heat will be enough to generate high-quality steam. Fig. 14 shows a comparison plot of the maximum temperature recorded for the three tested scenarios.

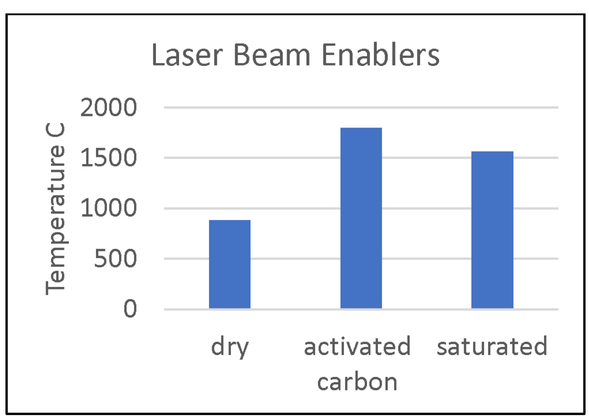

Figure 14—Summary of the maximum temperature results for each tested scenario.

Discussion

High-power laser technology has been proven in the field to be a sustainable and effective solution in various applications. The interaction between the laser beam and the rock causes a sudden increase in temperature, leading to a rapid expansion of the rock, which in turn creates fractures, mineral dehydration, and collapse, ultimately weakening the formation. To enhance the efficiency of the operation, enabling materials (e.g. activated carbon) have been identified such that, when exposed to the laser beam, they enable the rock to undergo a significant temperature increase. A novel approach has been developed to combine these enabling materials with the laser in a specialized tool. For field application, the tool can be connected to a high-power laser source stationed at the surface via an optical fiber transmission line, which can be injected in a coiled tubing, that can span a long distance. This specialized tool houses both a laser emitter and a nozzle that has the capability to inject the enabling materials, which can be in the form of powder or grains, onto the surface of the formation, concurrently with the laser beam. This synergy is expected to improve the heating of the rock, achieving higher temperatures with the same beam and power source. Experimental results have demonstrated the efficacy of this approach. Experimental findings show that when the laser interacts with dry rock, the temperature increases to 880 °C. However, when the rock is exposed to enabling materials, the temperature rises to 1795 °C, and when the rock is saturated, the temperature reaches 1564 °C. These findings highlight the potential of combining high-power laser technology with enabling materials to enhance the efficiency and effectiveness of various downhole applications.

From a technical perspective, the ability to double or tripe the effective thermal absorption of the target rock surface offers clear advantages. Higher temperatures not only lead to more efficient perforation but also allow for potential micro-fracturing, contributing to enhanced permeability. Early experimental and simulation studies showed that laser heating creates and area of enhanced permeability around the perforated region by developing micro cracks (Batarseh et al. 2003; Graves and Bailo 2005; Alrashed 2022). This is particularly valuable in tight formations or unconventional reservoirs.

Summary

A novel approach has been developed to enhance the efficiency of the laser stimulation process. The approach utilizes a patented specialized tool that can be connected to a high-power laser generator via a fiber optical transmission cable. The modular system allows for emitting the laser beam to the target sample. Enablers' materials such as activated carbon can be added to the sample to increase the efficiency of the laser stimulation process. To test the process, three scenarios were experimented and the following conclusions could be drawn:

- The presence of activated agent helps to improve the thermal absorption and elevates the temperature on the target surface.
- The high energy created as a result of adding the enabler material enhances the laser stimulation process by reducing the fracture pressure of the formation or creating micro cracks within the rock.
- Experimental findings show more than a two-fold increase in peak temperature on the target surface after adding the activated agent.
- The water saturated sample recorded a lower peak temperature than the dry sample after adding the activated agent, indicating that water absorbs the laser energy and reduces heat transfer to the formation.
- The recorded temperature on the water saturated sample (with activated carbon) is still high enough that the generated heat can produce high-quality steam.

References

Alrashed, A. 2022. Finite Element Analysis of the Laser Perforation Process: Thermal and Mechanical Perspectives. Ph.D. Dissertation, Colorado School of Mines, Golden, Colorado, United States. http://hdl.handle.net/11124/176636.

Alshunaifi, F. M., and Batarseh, S. I. 2022. High Power Laser-Enablers for Heating/Fracturing Stimulation Tool and Methods Therefor. US Patent Application Publication No. US 2022/0112772 A1.

Batarseh, S. 2001. Application of Laser Technology in the Oil and Gas Industry: An Analysis of High Power Laser-Rock Interaction and Its Effect on Altering Rock Properties and Behavior. Ph.D. Dissertation, Colorado School of Mines, Golden, Colorado, United States. http://hdl.handle.net/11124/78794.

Batarseh, S., Gahan, B. C., Graves, R. M., et al 2003. Well Perforation Using High-Power Lasers. Paper presented at the SPE Annual Technical Conference and Exhibition, Denver, Colorado, USA, 5-8 October. SPE-84418-MS. https://doi.org/10.2118/84418-MS.

Batarseh, S. I., Graves, R., San-Roman-Alerigi, D. P. et al 2017. Laser Perforation: Lab to the Field. Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 13-16 November. SPE-188729-MS. https://doi.org/10.2118/188729-MS.

Batarseh, S., San-Roman-Alerigi, D., Al Harith, A., et al 2019a. Laser Perforation: The Smart Completion. Paper presented at the Abu Dhabi International Petroleum Exhibition & Conference, Abu Dhabi, UAE, 11-14 November. SPE-197192-MS. https://doi.org/10.2118/197192-MS.

Batarseh, S., San-Roman-Alerigi, D., Al Obaid, O., et al 2019b. Laser Gun: The Next Perforation Technology. Paper presented at the SPE Middle East Oil and Gas Show and Conference, Manama, Bahrain, 18-21 March. SPE-194775-MS. https://doi.org/10.2118/194775-MS.

Graves, R. M., and Bailo, E. T. 2005. Analysis of Thermally Altered Rock Properties Using High-Power Laser Technology. Paper presented at the Canadian International Petroleum Conference, Calgary, Alberta, Canada, 7-9 June. PETSOC-2005-245. https://doi.org/10.2118/2005-245.